Lec-7

18/09/26 2:40 PM

2 | 39 — 1 ×10^0
2 | 19 — 1 ×10^1
2 | 9 — 1 ×10^2
2 | 4 — 0 ×10^3
2 | 2 — 0 ×10^4
2 | 1 — 1 ×10^5
0

```
while(n>0){
  rem = n%2
  sum=sum+rem×mul
  n= n/2
  mul=mul×10
}
```

```
while(n>0){
  rem = n% 10
  sum = sum + rem×mul
  mul = mul×2
  n = n/10
}
```

10 | 110101
10 | 11010
10 | 1101
10 | 110
10 | 11
10 | 1
0

1 ×2^0 → 1
0 ×2^1 →
1 ×2^2 → 4
×2^3 →
0 ×2^4 → 16
1 ×2^5 → 32
1

src → des

```
while ( n>0) {
int rem = n% des
sum =sum + rem×mul
n=n/des
mul = mul× src
}
```

1 0 1 0 1 1
2^5 2^4 2^3 2^2 2^1 2^0

Binary → octal

decimal

data Type

1
0

[--------] 8bit → 1 byte

①

data Type

Primitive
- byte
- short
- int
- long
- float
- double
- char
- boolean

non-primitive

byte → 8bit → 1byte
Short → 16 bit → 2byte
int → 32 bit → 4 byte
long → 64 bit → 8 byte

101

int x = 5
byte b = 5

0 0 0 0 0 101

$2^8 = 256$

_ _ _ _ _ _ _ _
2 2 2 2 2 2 2 2

101

sign-bit
-ve ← 1
+ve ← 0
MSB

1 1 1 1 1 1 1
1 1 1 1 1 1 1
LSB

[-127, 127]

1000 0000 → -0
0000 0000 → +0

1's Complement

00110110 → 54
11001001 → -54

2's complement

00110110 54
11001001
+1
11001010 → -54

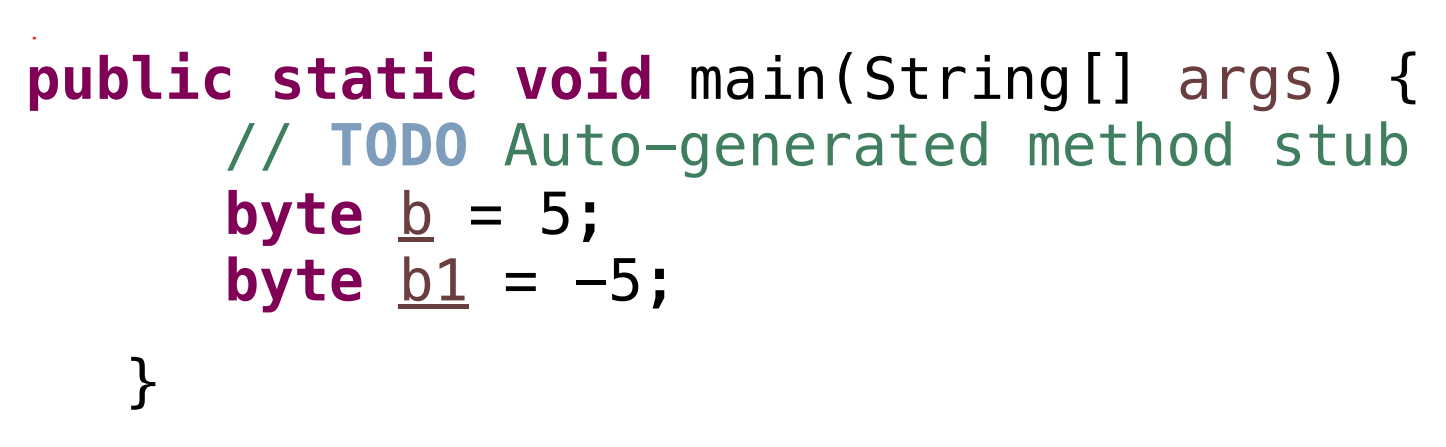
01101101 → x

10011101

01100010
+1
01100011

64+32+8+4+1=109

byte b = 5



```
public static void main(String[] args) {
	// TODO Auto-generated method stub
	byte b = 5;
	byte b1 = -5;
}
```

byte b=5

0 0 0 0 0 1 0 1

00000101
11111010
+1
11111011

byte

1 0 0 0 0 0 0 0
0 1 1 1 1 1 1 1

(-128, 127]

-128 to 127

1000 0000
0111 1111
+1
1000 0000

---- ----

byte → 8bit → -2^7 to 2^7-1
Short → 16bit → -2^{15} to $2^{15}-1$
int → 32 bit → -2^{31} to $2^{31}-1$ → 9digit
long → 64 bit → -2^{63} to $2^{63}-1$ → 18digit

420

1 1 0 1 0 0 1 0 0

10100100
01011011
+1 1 1
01011100 → -84

100101100

1 00101100

00101100

44